

MA2040 Jan-Apr 2022

Problem Sheet 5

1. Recall that the Moment Generating Function (MGF) $M_X(t)$ of a random variable t is defined to be $M_X(t) = E(e^{tX})$. Prove that:
 - (a) $M_X(0) = 1$.
 - (b) $\frac{d}{dt}(M_X(t))|_{t=0} = E(X)$. (c) $\frac{d^n}{dt^n}(M_X(t))|_{t=0} = E(X^n)$.
2. Let $X = \sum_{i=1}^n a_i X_i$ where $X_1, X_2, \dots, X_n$ are independent random variables and $a_1, a_2, \dots, a_n$ are constants. Prove that $M_X(t) = M_{X_1}(a_1 t) M_{X_2}(a_2 t) \cdots M_{X_n}(a_n t)$.
3. Compute the MGF for Binomial, Poisson, Exponential and Normal random variables.
4. Consider a fair coin with probability $p = 0.5$ of landing heads up and $q = 0.5$ of landing tails up. Let $X_1, X_2, \dots, X_n$ be independent random variables representing the outcomes of n consecutive coin flips.
 - (a) Define a random variable S_n as the sample mean of the outcomes of the first n coin flips, i.e., $S_n = \frac{1}{n} \sum_{i=1}^n X_i$. Show that $\lim_{n \rightarrow \infty} P(|S_n - p| < \epsilon) = 1$ for any $\epsilon > 0$.
 - (b) Suppose you conduct an experiment flipping the coin $n = 1000$ times and calculate the sample mean S_n . Using the result from part (a), estimate the probability that the sample mean deviates from the expected value $p = 0.5$ by more than $\epsilon = 0.05$.
5. Consider a fair six-sided die, with each face numbered from 1 to 6. Let $X_1, X_2, \dots, X_n$ be independent random variables representing the outcomes of n consecutive rolls of the die.
 - (a) Define a random variable S_n as the sample mean of the outcomes of the first n die rolls, i.e., $S_n = \frac{1}{n} \sum_{i=1}^n X_i$. Show that $\lim_{n \rightarrow \infty} \Pr(|S_n - \mu| < \epsilon) = 1$ for any $\epsilon > 0$, where μ is the expected value of a single roll of the die.
 - (b) Suppose you roll the die $n = 500$ times and calculate the sample mean S_n . Using the result from part (a), estimate the probability that the sample mean deviates from the expected value μ by more than $\epsilon = 0.5$.

6. Let $X_1, X_2, \dots, X_n$ be independent random variables, each following an exponential distribution with parameter λ , i.e., $X_i \sim \text{Exp}(\lambda)$ for $i = 1, 2, \dots, n$. Suppose you are given a random sample of size $n = 100$ from an exponential distribution with $\lambda = 0.5$. Use the central limit theorem to approximate the probability that the sample mean S_n falls within one standard deviation of its expected value.
7. Let $X_1, X_2, \dots, X_n$ be independent random variables, each following a uniform distribution on the interval $[0, 1]$. Suppose you are given a random sample of size $n = 50$ from a uniform distribution on the interval $[0, 1]$. Use the central limit theorem to approximate the probability that the sample mean S_n falls within two standard deviations of its expected value.